Sovannratana Khek

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SUMMARY

Computer science graduate student (B.S./M.S., Dec 2026) focused on backend and systems engineering. Builds RESTful APIs and full-stack apps, has run benchmark-driven vector search (HNSW) research, and has led teams as a program manager. Seeking a new grad software engineering role.

EDUCATION

Illinois Institute of Technology — Chicago, IL — *Expected: December 2026*
Combined B.S. & M.S. in Computer Science (4+1 co-terminal) | GPA: 3.2 | Dean's List, Fall 2024
Coursework: Data Structures & Algorithms, Database Organization, Operating Systems, Parallel Programming, Cryptography, Data Communications, Software Engineering, OOP

MIT Data Science & AI Program (online) — *In progress, Fall 2026*

EXPERIENCE

Program Manager | Exelon Summer Institute / Illinois Institute of Technology | Chicago, IL | *Summer 2026*

- One of two managers responsible for building, scheduling and supervising the summer's team of TAs and Leads, and for tracking housing, wage and activity budgets.
- Planned and ran the program's major events, including a 2-day AWS hackathon, a ComEd field trip and faculty talks; trained first-time TAs on curriculum I had authored the year before.

Computer Science TA → Lead | Exelon Summer Institute / Illinois Institute of Technology | Chicago, IL | *Summers 2023–2025*

- Built a Python-based in-browser coding environment so students could run assignments without a local setup, removing a common onboarding barrier for 20+ students per cohort.
- Designed course materials each summer, including slides, assignments, a Jeopardy-style review game and a digital escape room final assessment; as Lead (2024–2025), coordinated TAs and budgets.

PROJECTS

HNSW Insertion Order Research (Graduate Research — Python, CS 546) | *Spring 2026*

- Benchmarked five vector insertion strategies (random, K-Means, Hilbert curve, LID, and an original K-Means density approximation) on two BEIR datasets under two ef_construction settings, mentored by a PhD researcher.
- Measured graph quality with Recall@10 against an exact brute-force index, plus NDCG@10 and build, preprocessing and retrieval time; automated experiment runs with batch scripts.
- Found random ordering strongest overall; my density approximation matched it on Recall@10 without the $O(n^2)$ cost of exact LID; published findings in a written research report.

Oracle Lens — AI Coaching App (Rift Rewind Hackathon) | *Nov 2025 – Dec 2025*

- Led a 3-person team building an AI coach that analyzes a player's first 10 minutes of League of Legends gameplay and delivers role-aware feedback.
- Built the RESTful Flask (Python) backend integrating the Riot API with Claude 3.5 Sonnet on AWS Bedrock; deployed the MVP to Render and demoed it at the hackathon.

Greenhouse Temperature Simulation (Full-Stack Team Project — Python, FastAPI, React, MongoDB) | *Fall 2025 – Present*

- Led backend development for a 4-person team's passive solar greenhouse design tool, connecting a React simulator and dashboard to MongoDB; pairs a Python thermal physics model with an in-progress ML model for inverse design.

Console-Themed Portfolio Site (React, Vite, Vercel) | *Fall 2026*

- Built a data-driven portfolio where each project runs as a scripted demo in an on-page terminal; skill-to-project links are derived from project data so they never drift out of date.

Self-Correcting Compression Pipeline (Python) | *Fall 2025*

- Chained Huffman compression into a Hamming error-correcting code so messages survive single-bit transmission errors with no data loss; validated with scripted test cases.

C++ 2D Game Engine (Personal Project) | *Dec 2025 – Present*

- Building rendering, entity-component, game loop and input systems in C++ with raylib (migrating to OpenGL), using custom memory management and CMake for cross-platform builds.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, C#, SQL, HTML/CSS
Frameworks & Libraries: Flask, FastAPI, Django, React, Vite, Node.js, NumPy, MPI, RESTful APIs
Cloud & DevOps: AWS (Bedrock), Vercel, Render, Git, CMake, Bash, Batch
Databases & OS: PostgreSQL, MySQL, MongoDB | Linux (WSL), Windows